

Ben Irish

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EDUCATION

University of California, Los Angeles

Bachelor of Science in Statistics and Data Science, Minor in Systems Biology

Los Angeles, CA

Expected Jun 2029

- Current GPA: 3.9/4.0
- Relevant coursework: Statistics 13 and 20, Math 33A (in progress), Life Science 7A, Environment 10, EPS SCI 15
- Transfer coursework (UMass Global): Calculus II, Calculus III (in progress)

PROJECTS & WORK EXPERIENCE

Breast Cancer Subtyping from Gene Expression (*R*, *limma-voom*)

Los Angeles, CA

Independent Data Analyst

Apr 2026 – Present

- Processed ~1,200 TCGA breast cancer RNA-seq samples, combining limma-voom differential expression with hierarchical clustering of 2,000 high-variance protein-coding genes to test recovery of the four PAM50 subtypes
- Demonstrated that clustering isolated 185 of 190 Basal tumors but blurred the luminal subtypes (ARI 0.35), and that lncRNAs alone recovered subtype structure equally well while grouping individual luminal tumors differently
- Built a public Quarto report on a reproducible, version-controlled pipeline, presenting lncRNAs as a candidate source of breast cancer subtype markers while documenting the limits of a single-cohort, unsupervised analysis

Lyon and McAleer Creek Water Quality (*R*, *Quarto*)

Seattle, WA

Data Analyst, Lake Forest Park Streamkeepers

Jul 2026 – Sep 2026

- Analyzed 30+ years of King County stream monitoring data for two urban creeks, including water temperature, dissolved oxygen, and 198 storm and baseflow dissolved metals samples against state salmon habitat criteria
- Identified a ~1.4 °C step increase in summer temperature after 2013, a dissolved oxygen decline leaving half of Lyon Creek's readings below criterion, and storm copper above baseflow levels in 99% of sample comparisons
- Delivered a public report to the Lake Forest Park Streamkeepers to add to their website to show local residents how their creeks' health has changed, including a recommendation to log on-site rainfall in future storm sampling

Urban Coyotes in California (*R*, *sf*, *Quarto*)

Los Angeles, CA

Independent Data Analyst

May 2026 – Jun 2026

- Classified ~24,000 GBIF coyote observations across California as urban or non-urban using U.S. Census urban-area boundaries, filtered from 30,000+ raw records, to compare patterns across geography, time, and season
- Traced the urban share of observations from ~20% (2010) to ~50% (early 2020s), with urban sightings peaking in summer and non-urban in winter, weighing real urban colonization against the growth of citizen-science apps
- Published a public Quarto report on GitHub Pages cautioning wildlife-data users that rising urban coyote sightings partly reflect observer behavior, not just population growth, with rebuildable figures from the original GBIF DOI

LEADERSHIP EXPERIENCE

The Navigators

Los Angeles, CA

Program Member

Sep 2025 – Present

- Selected for Doulos, a year-long leadership and service development program, after a year of active involvement
- Lead Bible studies, worship, and one-on-one meetings to build community and support members' spiritual growth
- Planned events and attended weekly leadership meetings to coordinate ministry activities and member engagement

Shorewood High School – Associated Student Body

Shoreline, WA

Student Body Secretary/Vice President

Sep 2023 – Jun 2025

- Planned student council meetings and school-wide dances, and MCed assemblies for 1,500+ students and staff
- Represented Shorewood High School at district presidents' meetings and statewide ASB leadership conventions
- Spoke at class of 2025 graduation and directed the school Lip Dub, coordinating dozens of student performers

Wilderness Awareness School

Kenmore, WA

Volunteer Camp Counselor

Jun 2021 – Aug 2022

- Taught hands-on plant identification, wilderness survival, and sustainable camping practices to young campers
- Guided and supervised campers ages 5–12, keeping groups safe, engaged, and supported throughout the camp day
- Collaborated with camp leadership and staff to plan games, lessons, storytelling sessions, and educational skits

ADDITIONAL INFORMATION

- Skills: R, Quarto, Git/GitHub, Public Speaking, Event Planning, Team Coordination
- Interests: Football (Seahawks Fan; Commissioner of 4 Fantasy Leagues), Guitar/Bass, *East of Eden*, Cycling